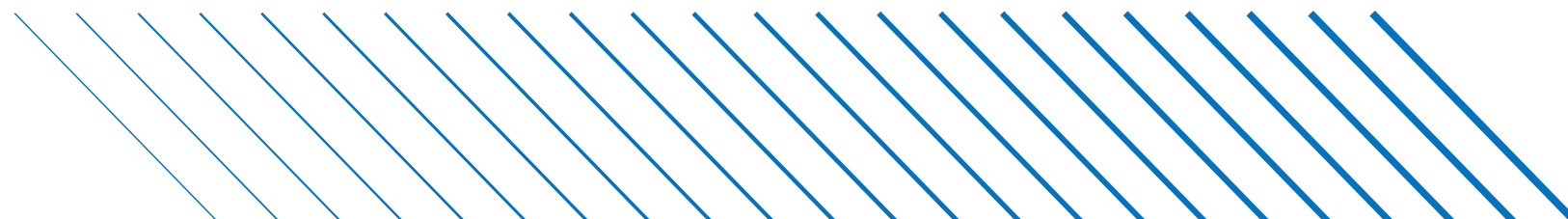
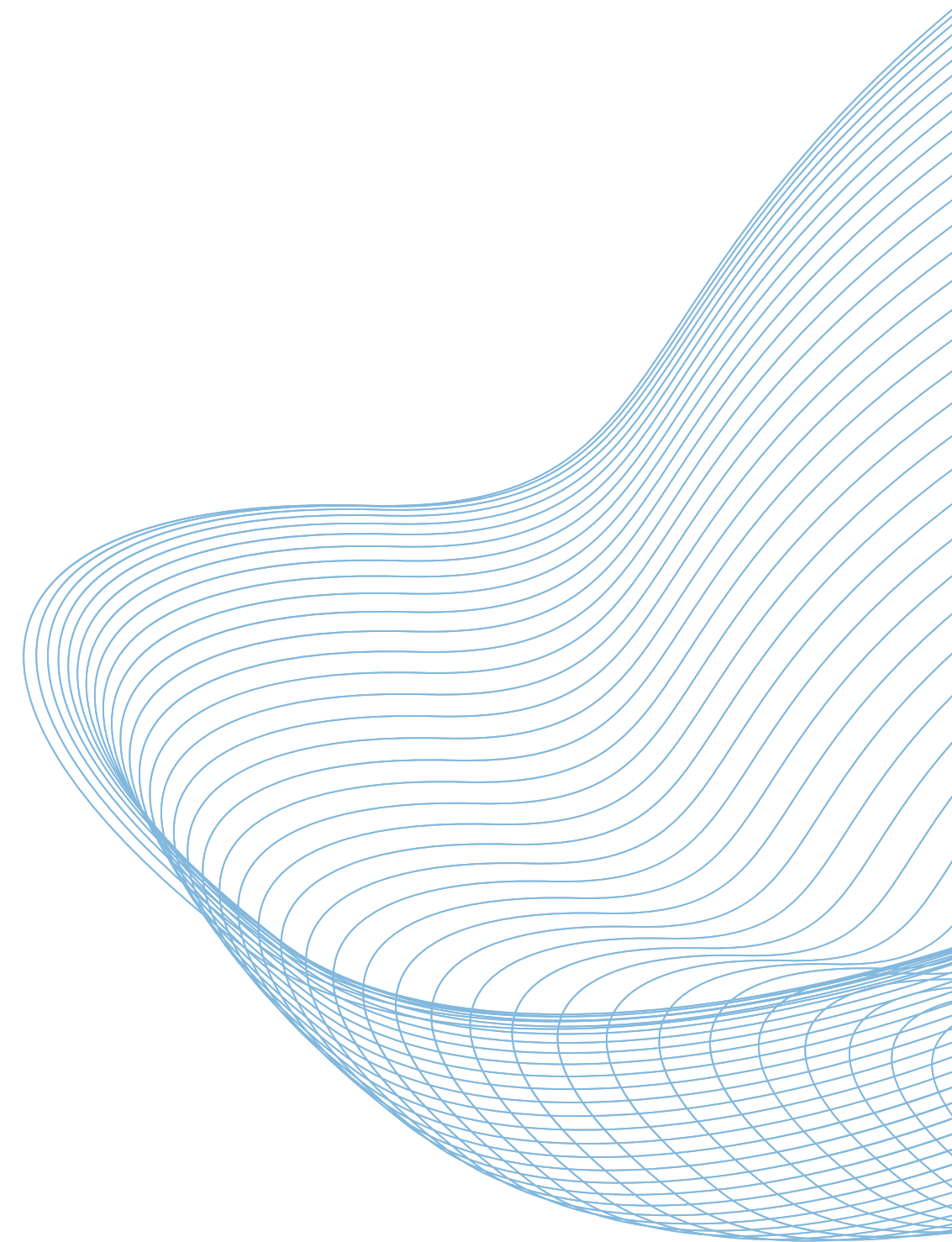


SageArk

Quant Training Session 40

Vol Surface Fitting



Overview of this Session

The purpose of this session is to understand what the volatility surface is and what is involved in a basic calibration.

What to Expect

Topics

- Topic 1: The Volatility Surface
- Topic 2: Properties of Vol Curves and Surfaces
- Topic 3: Model Types and Examples
- Topic 4: Calibration in Practice

At the end of this session, you should be able to **describe** tradeoffs between different types of models AND **calibrate** a volatility model to a snapshot of options market data.

US Equity Option Market

OCC Total Options and Futures Contract Volume					
28,434,668					
Options Download					
Total Options	Equity Options		Index/Other Options		
Exchange	Calls	Puts	P/C Ratio	Volume	Market Share
AMEX	808,809	855,186	1.06	1,663,995	5.89%
ARCA	1,857,733	1,760,381	0.95	3,618,114	12.8%
BATS	954,182	981,456	1.03	1,935,638	6.85%
BOX	950,625	839,010	0.88	1,789,635	6.33%
C2	584,969	577,139	0.99	1,162,108	4.11%
CBOE	2,417,825	2,450,492	1.01	4,868,317	17.22%
EDGX	939,368	869,074	0.93	1,808,442	6.4%
EMLD	432,456	348,065	0.8	780,521	2.76%
GEM	283,746	253,350	0.89	537,096	1.9%
ISE	646,789	616,134	0.95	1,262,923	4.47%
MCRY	215,545	197,283	0.92	412,828	1.46%
MIAX	722,317	668,871	0.93	1,391,188	4.92%
MPRL	570,318	560,317	0.98	1,130,635	4%
NOBO	433,693	503,057	1.16	936,750	3.31%
NSDQ	1,122,281	1,087,787	0.97	2,210,068	7.82%
PHLX	1,414,715	1,350,202	0.95	2,764,917	9.78%
Total	14,355,371	13,917,804	0.97	28,273,175	100%

OCC Data for July 25, 2022

OCC Total Options and Futures Contract Volume					
40,011,948					
Options Download					
Total Options ▼					
Exchange	Calls	Puts	P/C Ratio	Volume	Market Share
AMEX	1,540,495	1,526,247	0.99	3,066,742	7.69%
ARCA	2,493,991	1,785,489	0.72	4,279,480	10.74%
BATS	1,176,862	785,394	0.67	1,962,256	4.92%
BOX	1,525,353	1,086,155	0.71	2,611,508	6.55%
C2	844,433	751,426	0.89	1,595,859	4.00%
CBOE	4,075,697	3,340,982	0.82	7,416,679	18.61%
EDGX	1,137,953	951,025	0.84	2,088,978	5.24%
EMLD	651,100	403,225	0.62	1,054,325	2.64%
GEM	687,517	451,718	0.66	1,139,235	2.86%
ISE	1,115,607	939,742	0.84	2,055,349	5.16%
MCRY	441,274	310,946	0.7	752,220	1.89%
MIAX	1,243,457	816,418	0.66	2,059,875	5.17%
MPRL	1,305,144	861,720	0.66	2,166,864	5.44%
NOBO	808,751	716,790	0.89	1,525,541	3.83%
NSDQ	1,231,456	920,058	0.75	2,151,514	5.40%

OCC Data for July 24, 2023

US Equity Option Market

Data as of August 3, 2021 / Source: OCC, JAX Contract Servers

- **OCC lists over 5000 equity underlyings**
- **Hundreds of thousands of listed options:**
 - Calls and Puts
 - Many Strikes and Expirations
- **Single Name Examples**
 - AMZN ~6000 Options
 - GOOG+GOOGL ~10000 Options
 - JPM ~1000 Options
- **Option market prices move around as the underlying market moves**
- **How does an OMM make sense of all this?**



Topic 1: The Volatility Surface

The Volatility Surface: Implied Volatility

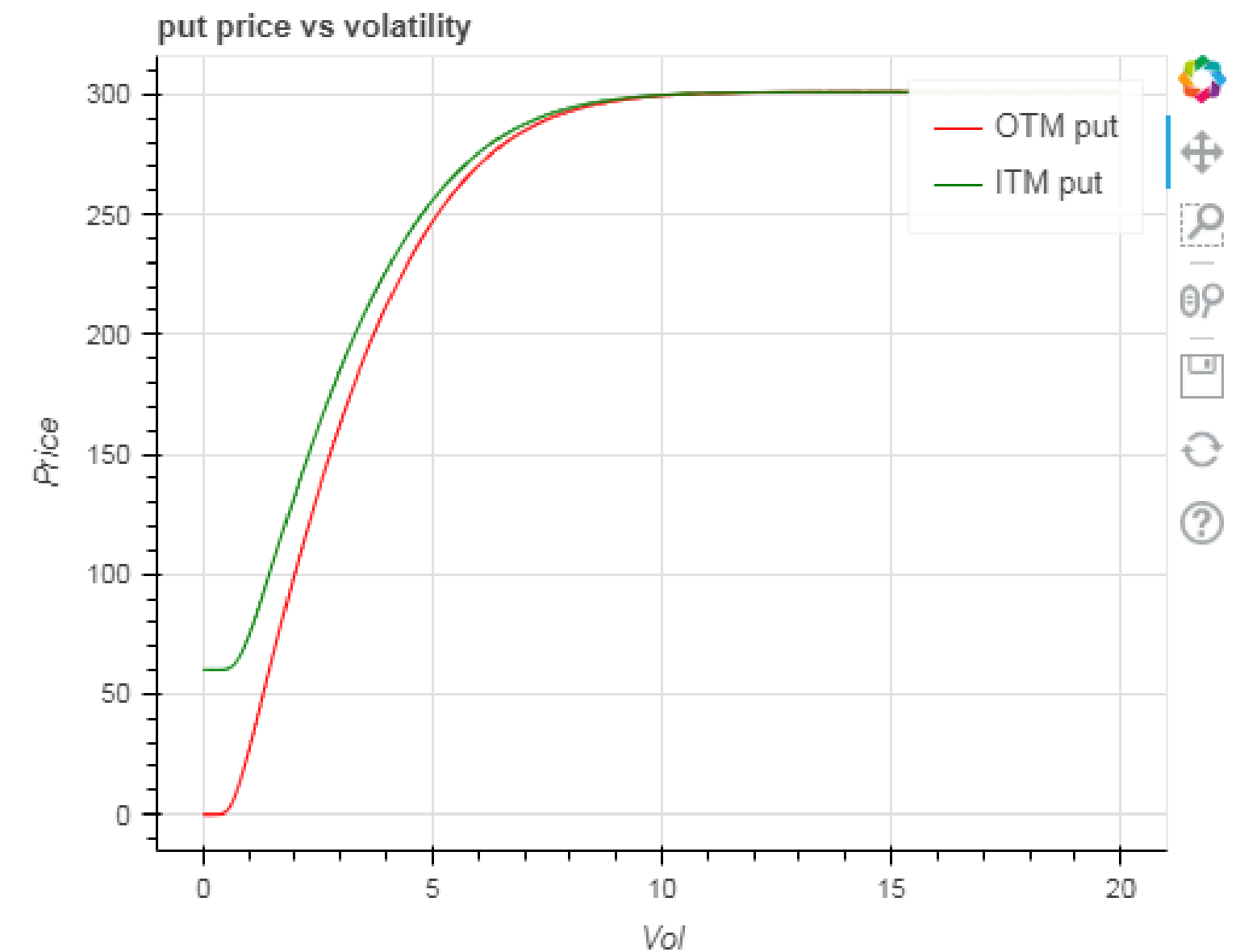
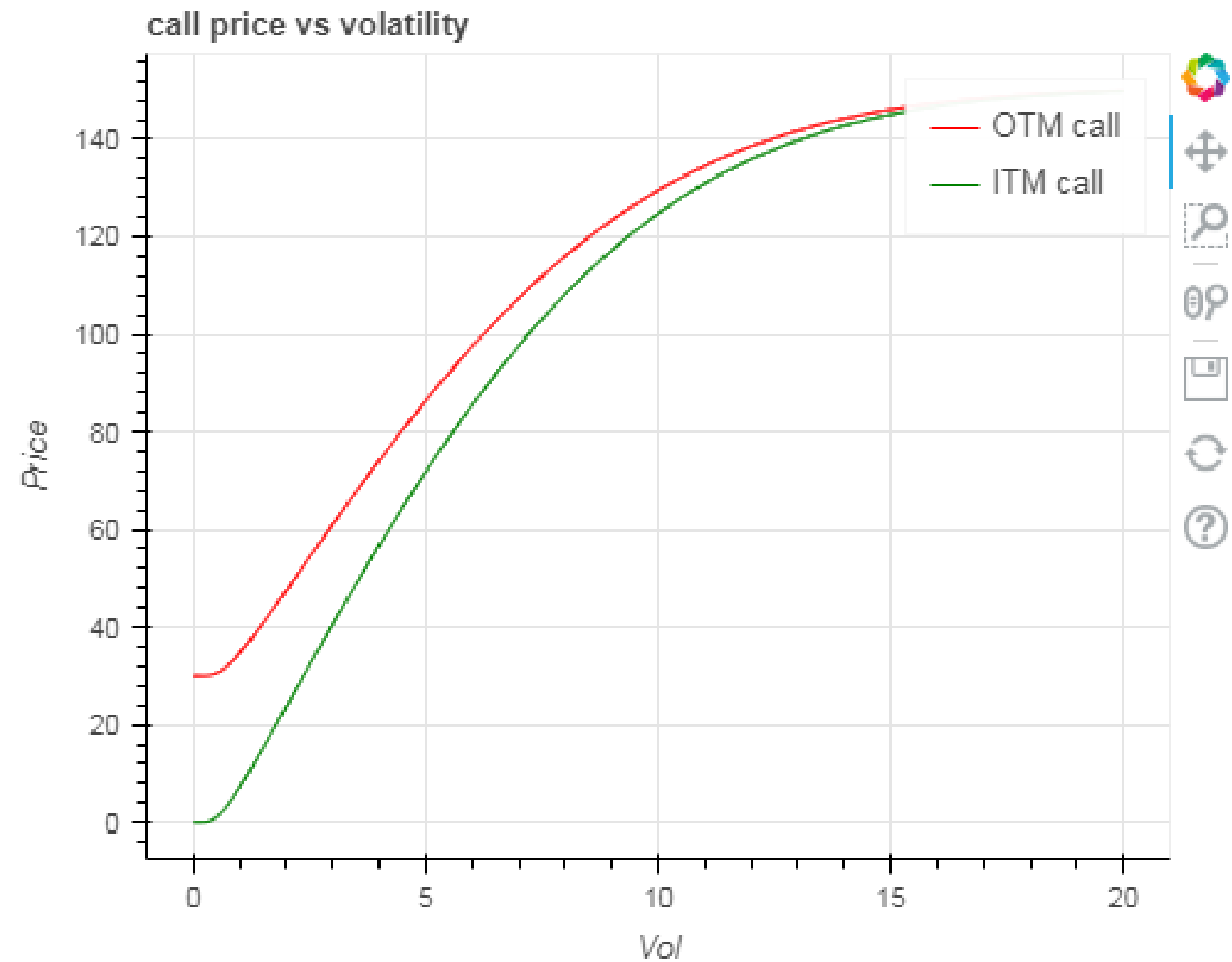
- Pretend that option prices arise from a log-normal underlying price distribution, per the Black-Scholes PDE, assuming all other inputs than volatility are known.
- Back out the volatility under this model that would reproduce the observed market price. This is the ***implied volatility*** or “**IV.**”

Reality

The pricing inputs other than volatility are not known exactly,
and neither is the market price of an option.



Implied Volatility



- Cannot compute IV **if the option price is too high** (e.g., when the put exceeds K or the call exceeds S for Americans).
- Cannot compute IV **if the option price is too low** (e.g., below “intrinsic value” for Americans).
- **If the target price is close to any bound**, the IV calculation is possible but typically not useful.

Volatility Surface

- The **vol surface** represents the implied volatility for options on a given underlying across strikes and expirations.
- The **3d plot** of the surface is fun to look at, but not especially useful.
- **Practitioners** generally use:
 - **Vol curves** (slices of the surface for a particular expiration)
 - **Term structure plots** (volatility or other quantities vs. time to expiration)

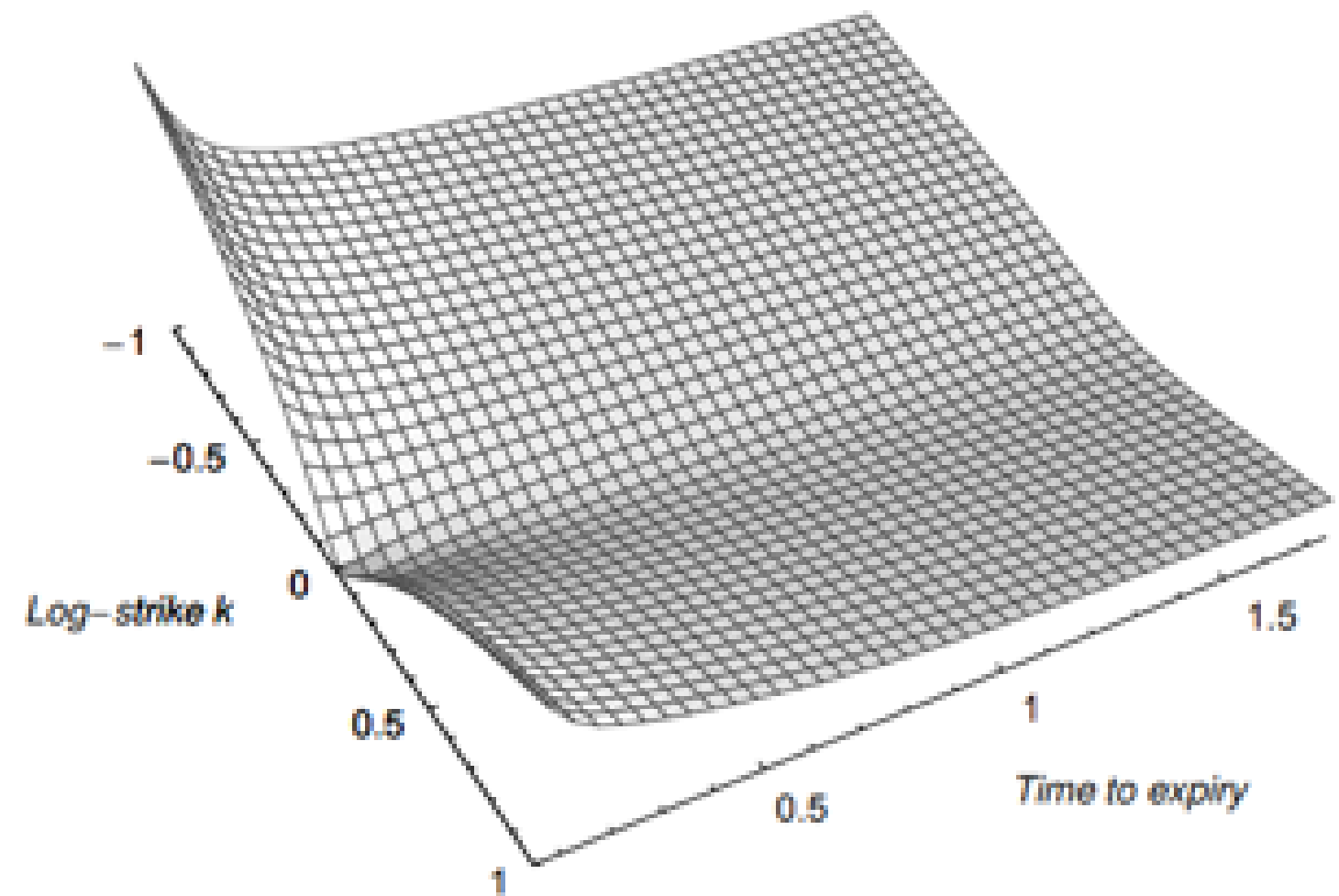


Figure 1: Graph of the SPX implied volatility surface as of the close on September 15, 2005. The day before triple witching.

<http://faculty.baruch.cuny.edu/jgatheral/ImpliedVolatilitySurface.pdf>

Class Question

Question

Is IV a vacuous concept?

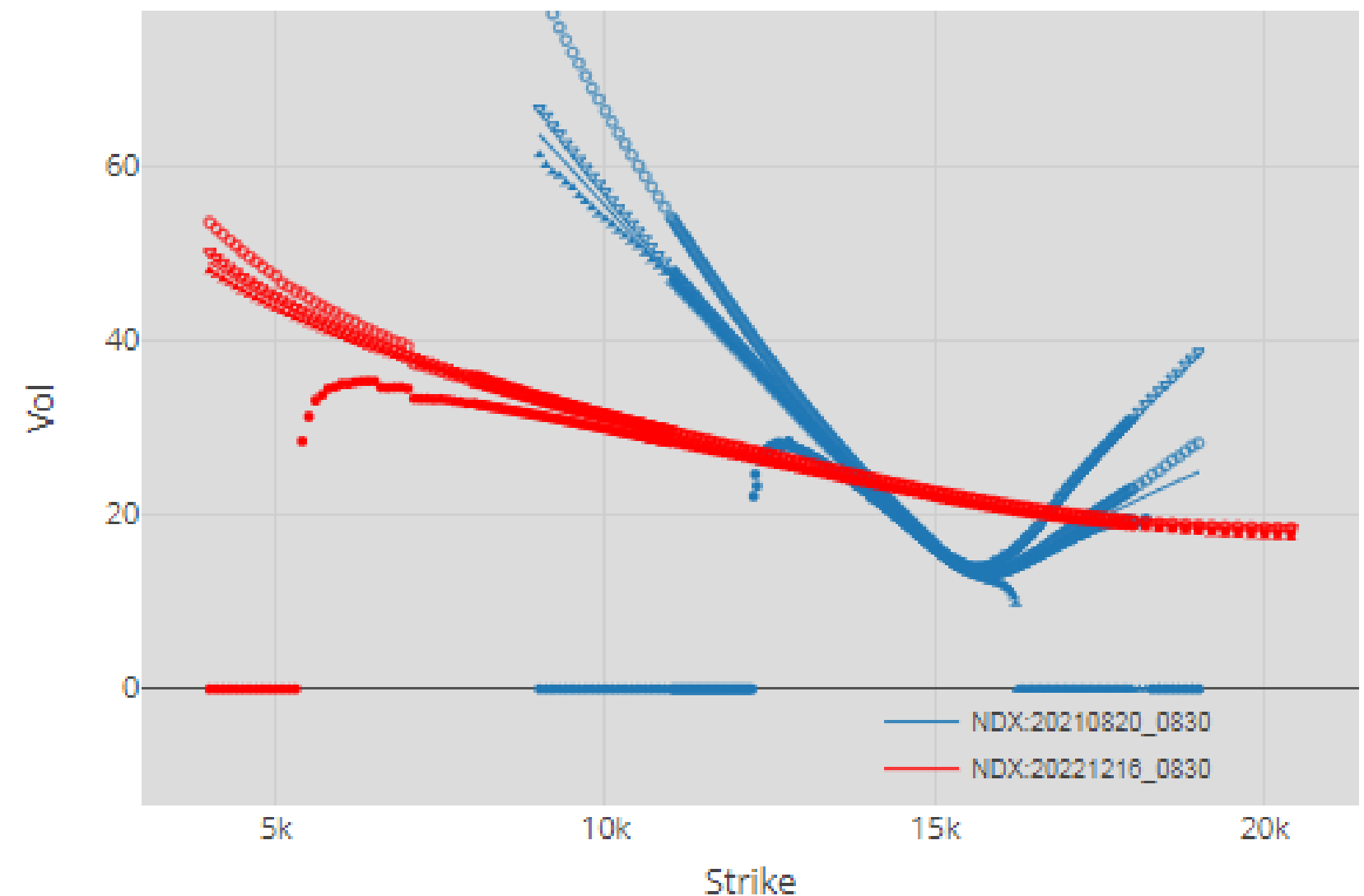
Answer

If we price an option by backing out an IV from its market price, then plugging that number back into the pricing formula – haven't I just expressed a pointless tautology?



Vol Curve Example: NASDAQ

- **There is not one market price because:**
 - At each strike, there is a call and put
 - There is a bid/ask for each option quote
- **Generally prefer the OTM option (puts at low strikes, calls at high strikes) because:**
 - More about vol, less about delta
 - Less impacted by uncertainty in the underlying
 - Less early-ex impact in the American case
 - Typically quoted tighter (in vol space)

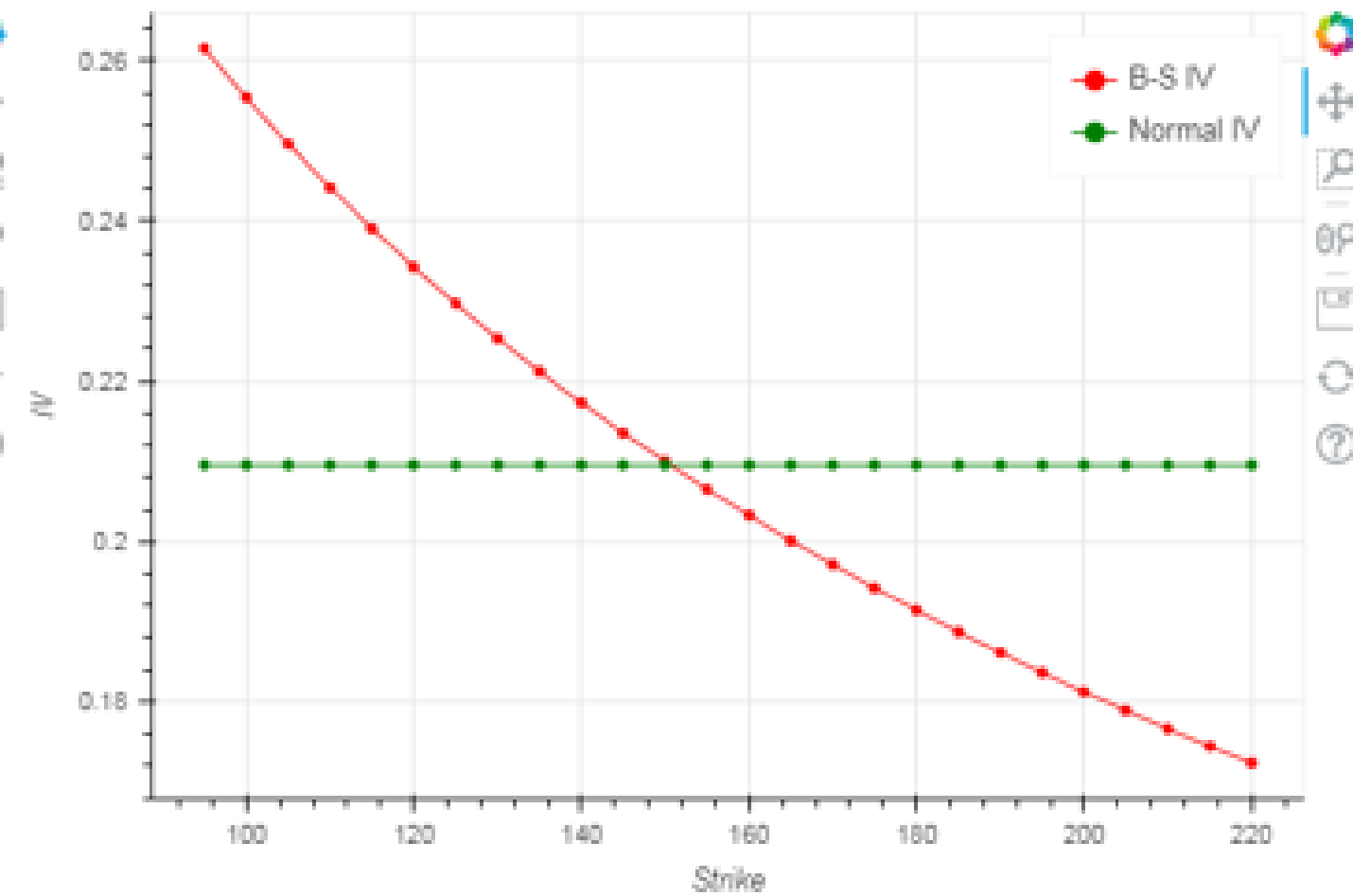
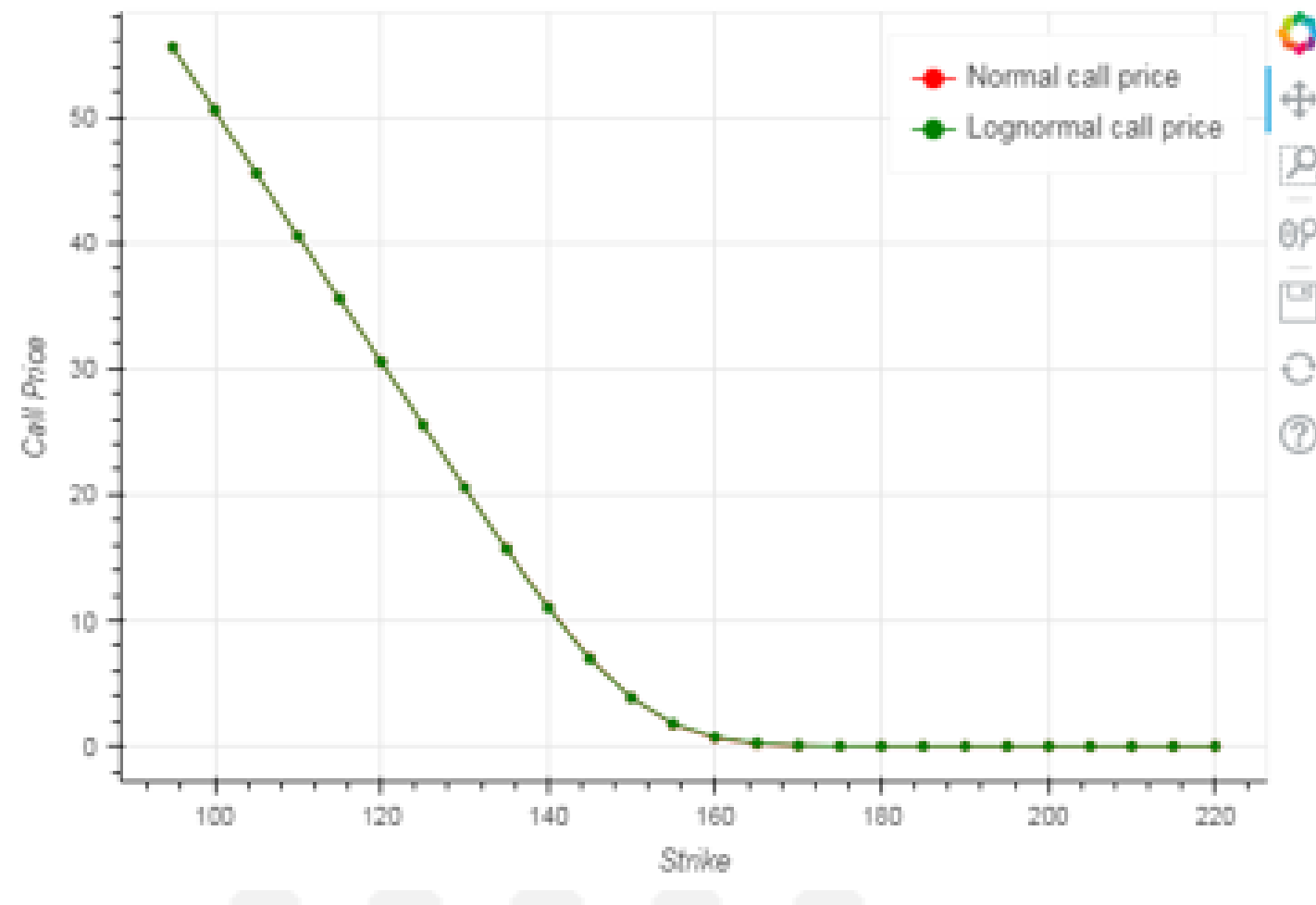


Digression: Normal Vol

- The **Bachelier Model**, based on a normal price process, pre-dated the Black-Scholes model based on the log-normal process.
- For **stock price modeling**, quants tend to disfavor the normal to avoid possible negative prices (though truncation is an option).
 - Some products can have negative prices – e.g., rates, and as we learned in April 2020, oil.
- A **normal process** may have some advantages – e.g., a normal will generate some skew when viewed through a lognormal lens.

Keep an open mind!

Normal Volatility



- Assume **stock price** is normally distributed, rather than returns
- Truncate **distribution** so there is a discrete mass at zero
- Applying **lognormal IV** to a normal reality yields a skew

Calibration

- **Calibration**

- It is the extraction of a volatility surface from market data
- It views the market as mostly true (with some admixture of noise and other distortions)

- **Questions**

- Why?
- Strawman: Why not just trade using market implied vols directly?



Market Example

Way OTM Options:

Situation

On Monday 7/26/2021, for the NDX options expiring two days later on Wednesday, the 3800 put market is (300) 0.05 / 0.10 (1262) and strikes between 1200 and 3780 are all (no bid) / 0.5.

In particular, the 1200 and 3500 strikes put offers are both 0.05 with size about 600 contracts.

Questions

- Should I value the 1200P and 3500P at the same price?
- What about their implied volatilities?



Why the Vol Surface?

- 1 **Theos:** More accurate **theos** by reducing noise in the raw market implied vols.
- 2 **Priors:** Express **priors** on the relationships between different options.
- 3 **Liquid Options:** Less **liquid options** are valued based on their relationships with more liquid options.
- 4 **Surface:** The **surface** varies relatively slowly, so we can use it even during periods of reduced liquidity.
- 5 **Dimensionality:** Reduce the **dimensionality** of market observations.
- 6 **Use Risk Factors:** Define useful **risk factors** in volatility space.
- 7 **Trade Spreads and Light Exotics:** Needed to **trade spreads and light exotics** effectively (e.g., VA, VIX).

Theoretical Value Accuracy

Theoretical Values

We usually need **theoretical values** (or “theos”) more accurate than afforded by the bid-ask spread.

Making Market S

Making Markets in option spreads typically requires more accurate theos than would be implied by combining uncertainties from the outright.

Exotics

Exotics also require accurate theos (e.g., variance swap integral).

Topic 2: Property of Vol Curves and Surfaces

Properties of the Vol Surface

Must Haves:

- No Static Arbitrage
- Stability
- Smoothness
- Fits Markets

Should Haves:

- Intelligibility
- Parsimony

No Static Arbitrage

- No Static Arbitrage Trades Available
 1. Butterflies
 2. Verticals
 3. Calendars
- Implied PDF Non-Negative
- Slopes are Bounded
- Asymptotic Behavior (Lee)
- Monotone Total Variance vs Time to Expiration

Asymptotic Behavior Bound

- **Lee 2003, “The Moment Formula for Implied Volatility at Extreme Strikes”**
 - Asymptotically
Example: As
 - Implied Variance cannot increase faster than linearly vs $\log(K)$
 - Implied Vol cannot increase faster than square-root log-moneyness
- **Systems typically assume this kicks in several stddevs away from ATM**

Stability

- If our **theos** fluctuate or oscillate around a value, an adversary can arb us by selling to us when our theo fluctuates high and buying from us, when it fluctuates low.
- Similar situation, if the **value is slowly changing** and we fluctuate or oscillate around the trend.
- Tradeoff **between stability and responsiveness** to genuine changes.
- **Algorithms** need to be stress-tested (back testing against normal, everyday markets may give a false sense of security).



Topic 3: Model Types and Examples

Vol Surface Model Types

Parametric Curve

Define a functional form, and fit the dots (e.g., Polynomial)

Non-Parametric or Semi-Parametric

Large number of parameters that can fit almost anything, with complexity reduced by regularization (e.g., Smoothing Spline)

Structural or Stochastic Process Model

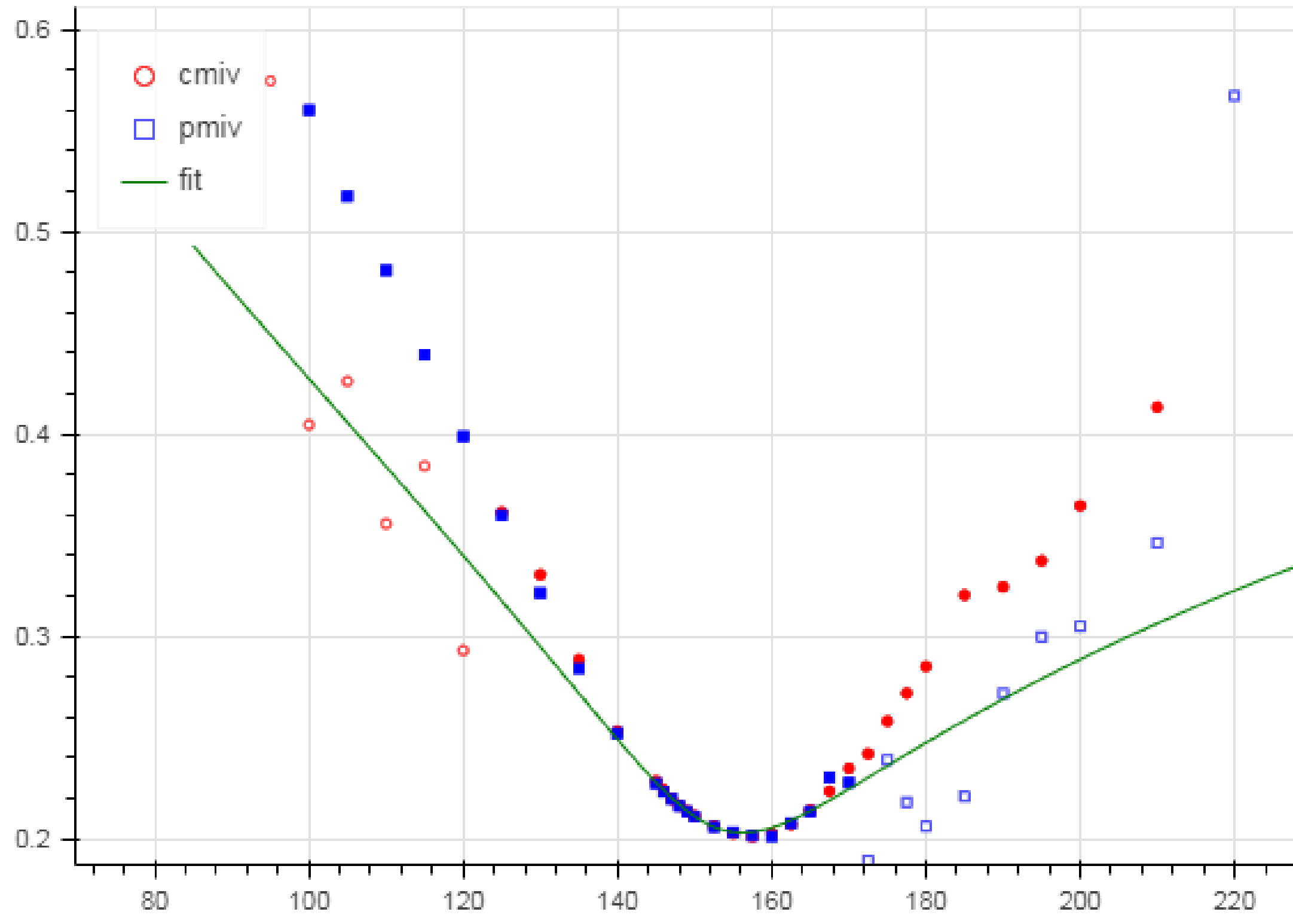
Define a process under which the system evolves (e.g., Heston Model or Local Volatility)

PDF-Based Fitting

Model the Terminal PDF, Map to Vol Curves by integrating

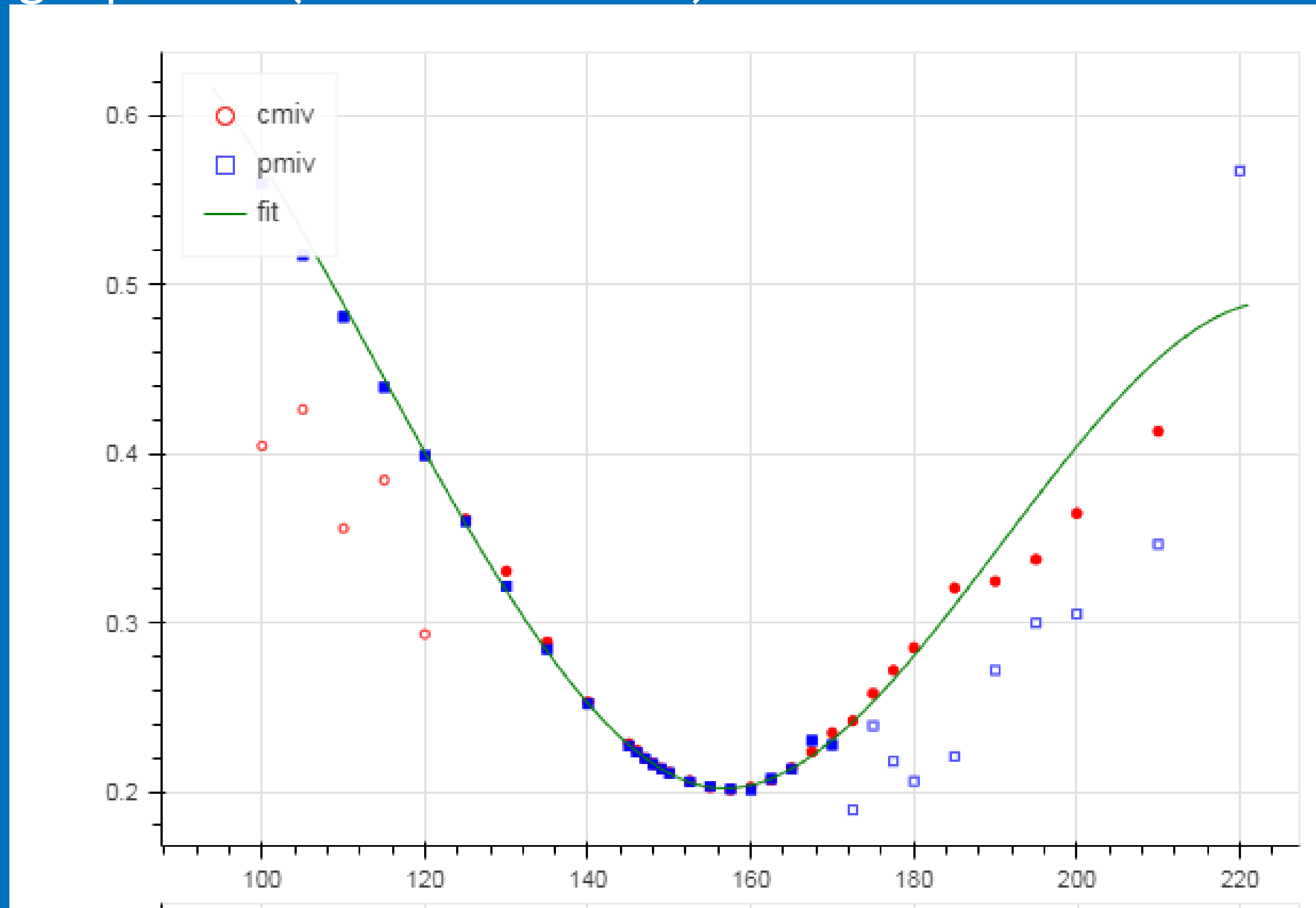
Parametric Fit Example

SSVI/S3



Non-Parametric Fit Example

Smoothing Spline (Off-the-Shelf)



Structural or Process Model Fits

Local Volatility

Only **one source** of randomness, but volatility is a function of time and spot price.

Stochastic Volatility

Examples Include:

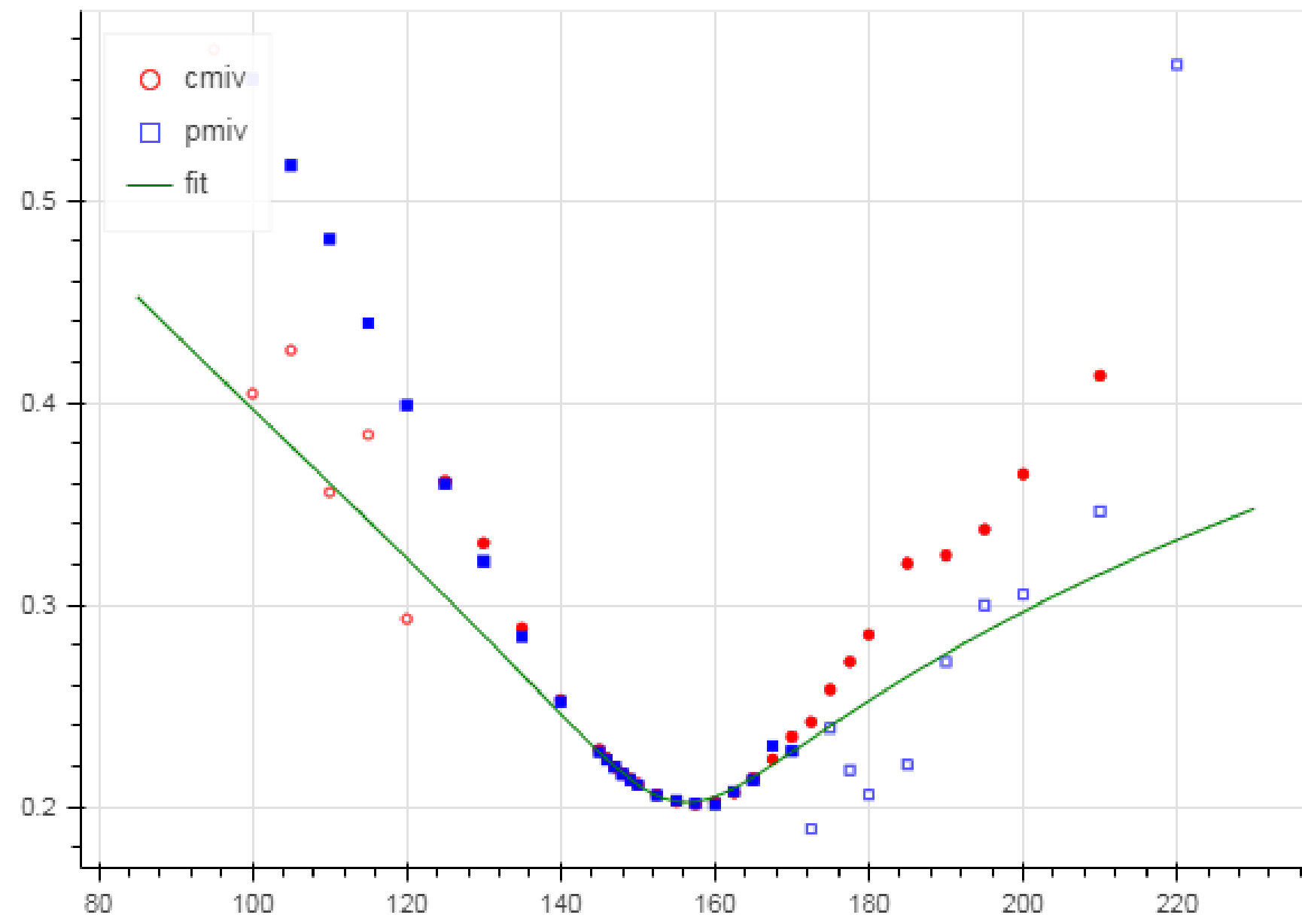
- Heston
- Kou
- Etc...

Stronger Statement

These models each make a **stronger statement** than a Curve or PDF Fit (e.g., they predict densities at any time before expiration).

Example

Heston Model Fit



- Heston Model fit to the JPM data
- Characteristic function based calculation (Credit: Ta-Wei Wang)
- Fit takes much longer than other methods

$$dS_t = rS_t dt + \sqrt{V_t} S_t dW_t^1$$

$$dV_t = a(\bar{V} - V_t)dt + \eta \sqrt{V_t} dW_t^2$$

$$dW_t^1 dW_t^2 = \rho dt$$

Topic 4: Calibration in Practice

Calibration in Practice

Input Data:

1. Current Market Bids and Offers for Options and Underlying
2. Trades
3. Related Markets (e.g., NDX and QQQ options)
4. Prerequisite Info: Rates, Times, Dividends
5. History
6. Priors and Constraints (e.g., reject a fit-over-fit move of more than X in parameter P - Bayesian Framework)

We will only discuss (1) in any detail.

The Axes

There are several possible choices.

Vol Curve Abscissa

- Strike: K
- Log-Strike: $\log(K)$
- Moneyness: $K - F$ or K/F
- Log-Moneyness: $\log(K/F)$
- Time-Scaled Log-Moneyness or tslm:
- Normalized Strike:
- “ z ” = (similarly, d_1 d_2)

Y-Axis

- Volatility
- Variance
- Log-Volatility



Targeting

- **Per-Option Targeting**

Either compute the IV for the bid and offer and combine, or compute a mid-price and corresponding IV. Example:

$$\text{targetIV} = (\text{aSize} * \text{bIV} + \text{bSize} * \text{aIV}) / (\text{aSize} + \text{bSize})$$

OR

$$\text{targetIV} = \text{IV}((\text{aSize} * \text{bid} + \text{bSize} * \text{ask}) / (\text{aSize} + \text{bSize}))$$

- **Per-Strike Targeting**

Combine the call and put for a given strike. (e.g., Can optionally use delta-vega based weights to eliminate uncertainty in the underlying to 1st order.)

- **One-Sided Targets**

One-sided targets on wings where the OTM options has no bid (cost function should penalize crossing).

Cost Function

- Typical starting point is least-squares
- However, one-sided targets or “repel from bid or offer” logic can introduce non-linearity
- Weights based on training priorities and vol information content

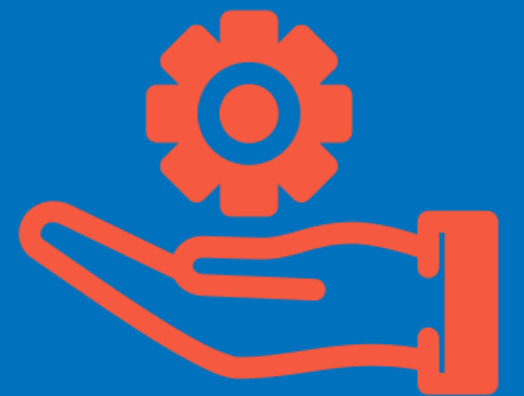
Examples:

1. A weight-factor proportional to vega V or to $1/(\mathbf{NS} + \mathbf{c})^p$ to emphasize the near-ATM region
 2. A weight factor proportional to $1-|\Delta|$ to deemphasize ITM options
 3. A factor varying inversely as the bid-offer spread to accord tighter quotes, higher weight
- Penalties to avoid arbitrage conditions, if necessary

Lots of variations!

Optimization

- Use off-the-shelf algorithms, don't reinvent the wheel!
- Decide term-by-term or all at once
- Initial state selection
- Constrained or unconstrained
- May need “reset” functionality for getting out of local minima traps



A Calibration! Now What?

We're not quite done yet, lots left to do!

- Are the fits noisy or stable? Tune regularizations.
- Track down any outliers or glitches
- Optimize performance
- Metrics and monitoring
- Model all the things...never ending variety!



(cf: W. Schoutens et al, “*A Perfect Calibration! Now What?*”)

Still, Now We Can...

- Make markets based on calibrated theoretical values
- Run takeout strategies
- Prop trade surface anomalies
- Price exotics and trade for edge
- Understand our vol surface risk via meaningful or at least parsimonious factors
- PnL
- And, last but not least...**Study More!**



THE END